FOR COMPLETE BEGINNERS

Free delivery?

Complete Modern Java, episode 10: decisions with if and else, in Java and TypeScript

> A comparison gives true or false. if and else run exactly one branch, then the program carries on.

Free delivery from $50. Sarah's basket is $52.40, kept as 5240 cents.

The program has to choose: free delivery, or $4.99.

In this episode

- A comparison is a boolean
- if and else
- The six comparison operators
- = versus ==
- TypeScript ===

Coming next

- and, or, not, else if (episode 11)
- Loops (episode 12)
- Methods (episode 13)

At the time of recording: Java 25, Node 24 LTS, IntelliJ IDEA 2026.1.3.

01 What each line printed

totalCents >= 5000 (5240)	`true, Free delivery, Thank you, Sarah`
totalCents >= 5000 (4240)	`false, Delivery: $4.99, Thank you, Sarah`
items == 3 items != 3	`true false`
5000 > 5000 5000 >= 5000	`false true`
if (items = 5)	`error: int cannot be converted to boolean`
TypeScript items === 3	`true`

Every row was run, not predicted.

02 if and else

The condition becomes true or false. Exactly one branch runs, then the program carries on after it.

>= means greater than or equal to, so exactly 5000 cents qualifies. One = assigns; two == compare.

Delivery.java

```
int totalCents = 5240;
if (totalCents >= 5000) {
    IO.println("Free delivery");
} else {
    IO.println("Delivery: $4.99");
}
IO.println("Thank you, Sarah");
```

> Printed true, Free delivery, Thank you, Sarah; with 4240: false, Delivery: $4.99, Thank you, Sarah.
>
> Run on a Mac, 2026-10-24: Java 25.0.4.1, Node 24.22.1, TypeScript 7.0.2.

03 TypeScript

The same if and else. Use === and !==: JavaScript's == can convert values before comparing, === never does.

delivery.ts

```
const items: number = 3;
console.log(items === 3);   // true
```

> npx tsc --noEmit was quiet; node printed Free delivery, Thank you, Sarah, true.
>
> Run on a Mac, 2026-10-24: Java 25.0.4.1, Node 24.22.1, TypeScript 7.0.2.

Write a decision

1. Write the rule in words first: from $50, free delivery.
2. Pick the sign that matches it: >= includes exactly $50.
3. One = assigns. == compares. In TypeScript, ===.
4. Remember the line after the if and else runs either way.
5. Challenge: predict 4999, 5000 and 5001 cents.

Everything else

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- **The code and the challenge files: complete-modern-java**
  https://github.com/code-with-sam-dev/complete-modern-java

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